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(54) **HAIR BRAIDING DEVICE**

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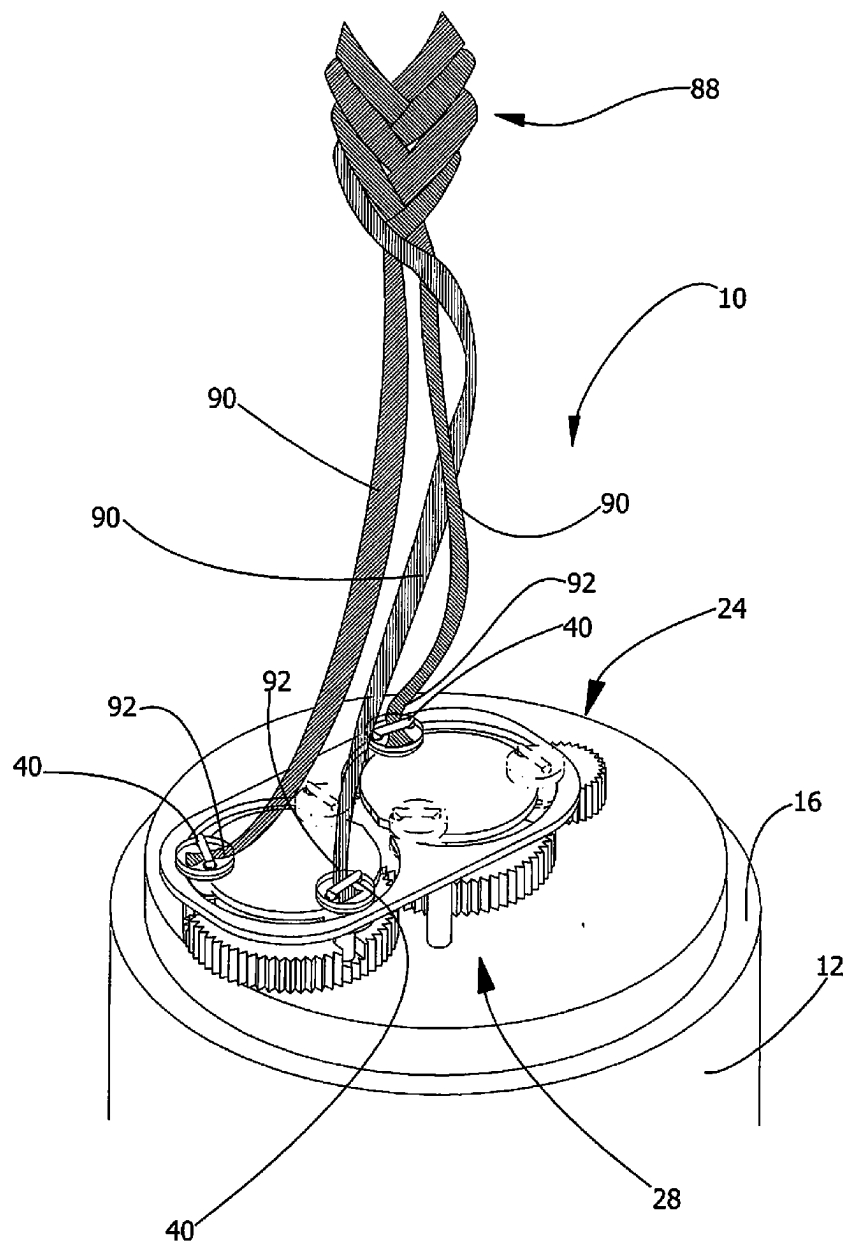
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CPC **A45D 2/00** (2013.01); **D04C 3/00**
(2013.01); **A45D 2002/005** (2013.01)

(57)

ABSTRACT

A hair braiding device for automatically braiding hair includes a housing which defines an orifice that extends into the housing. A motor is mounted in the housing and is operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction. A braiding mechanism is operably coupled to the output shaft of the motor and is mounted in the orifice of the housing. The braiding mechanism comprises a plurality of hair couplers which are configured to move along a figure-eight path when the motor is activated.



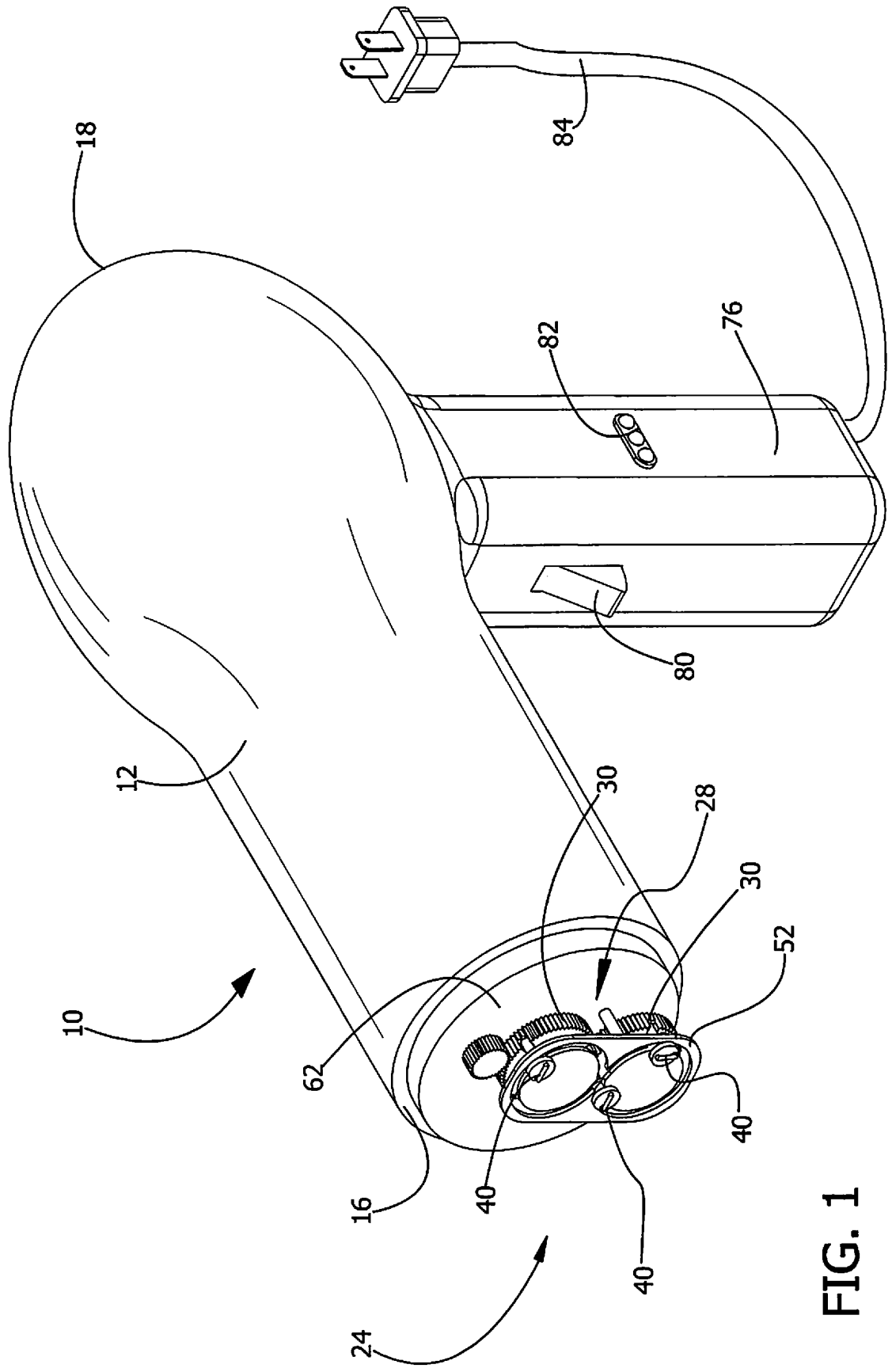


FIG. 1

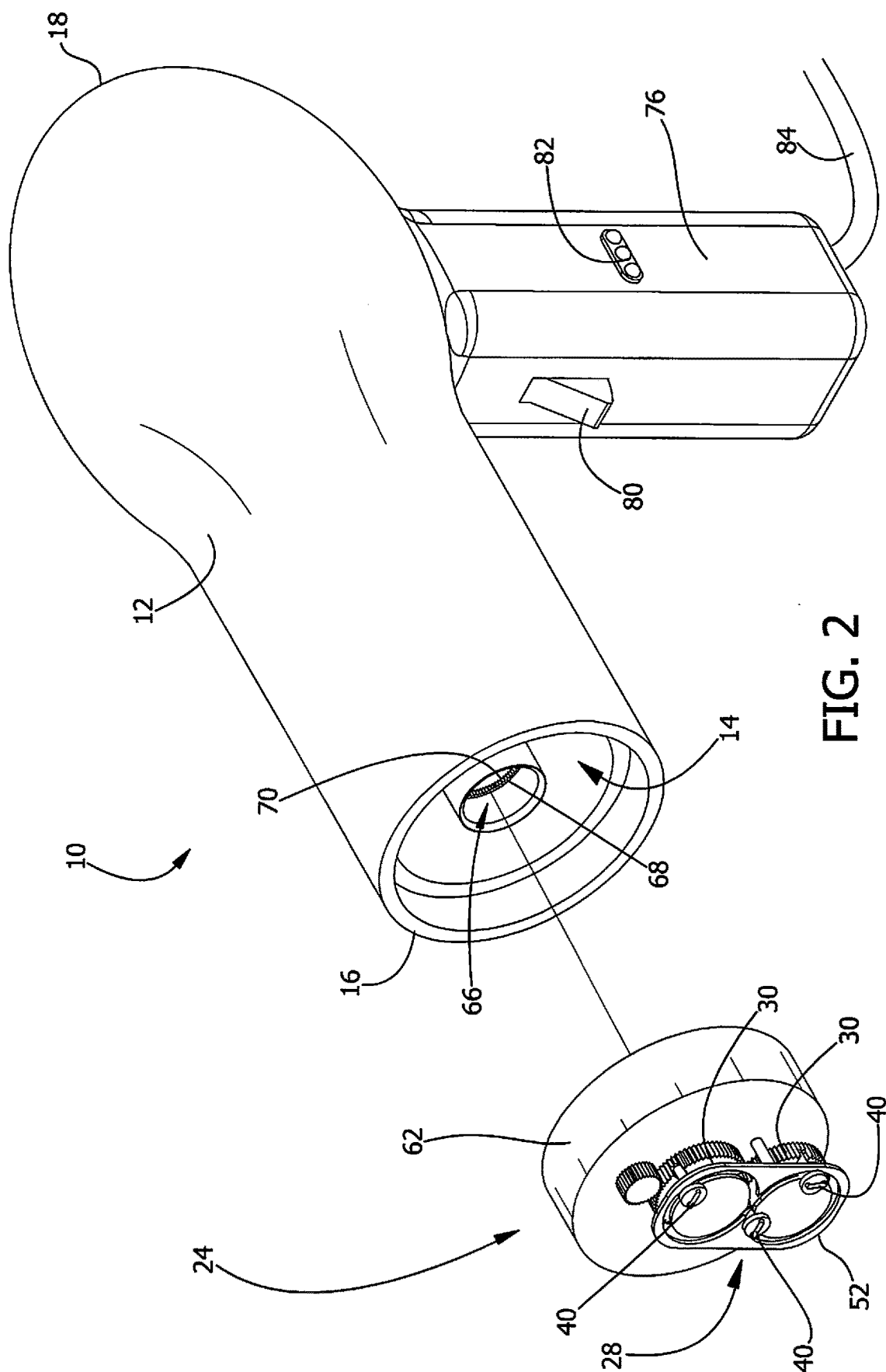


FIG. 2

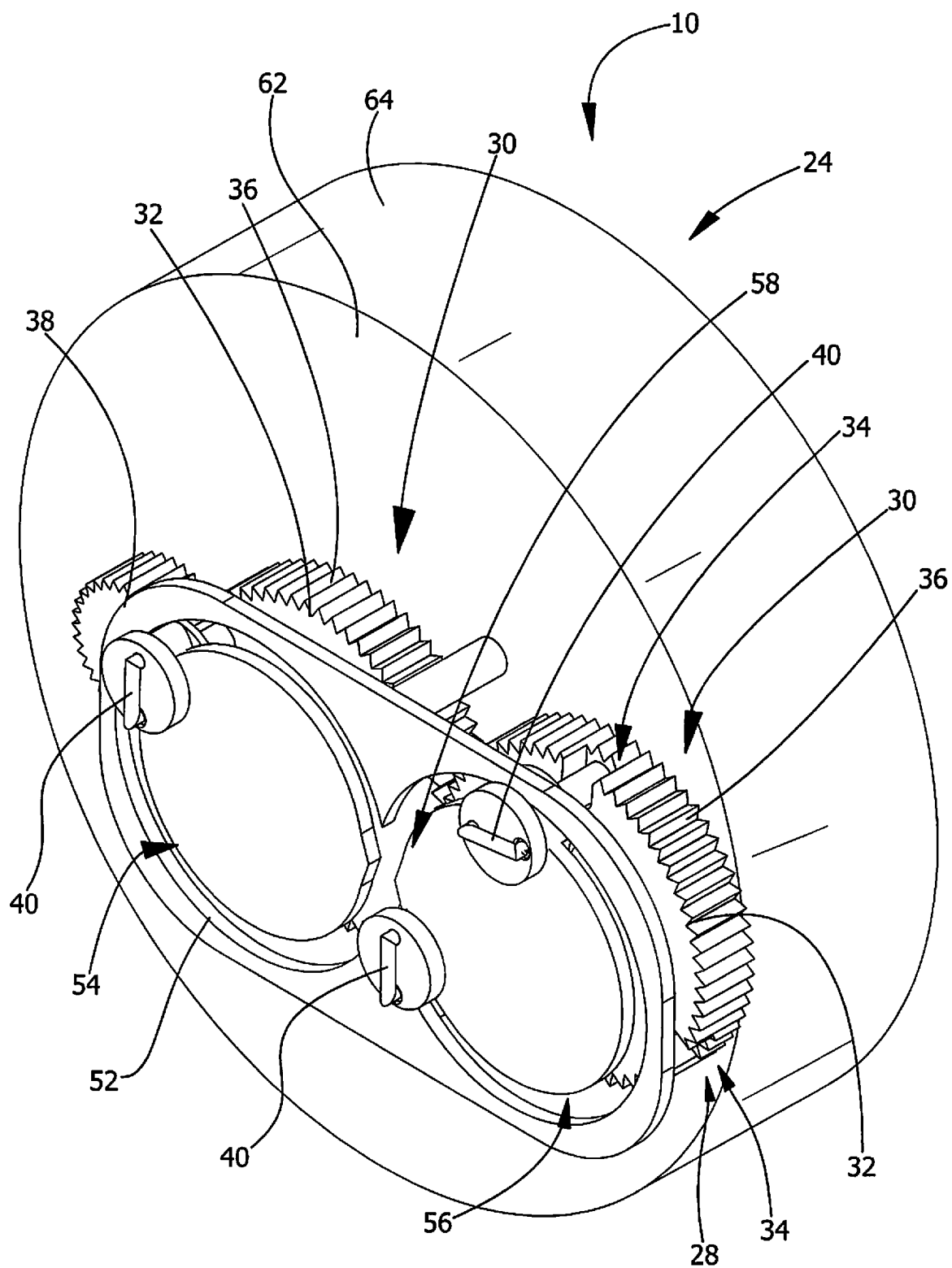


FIG. 3

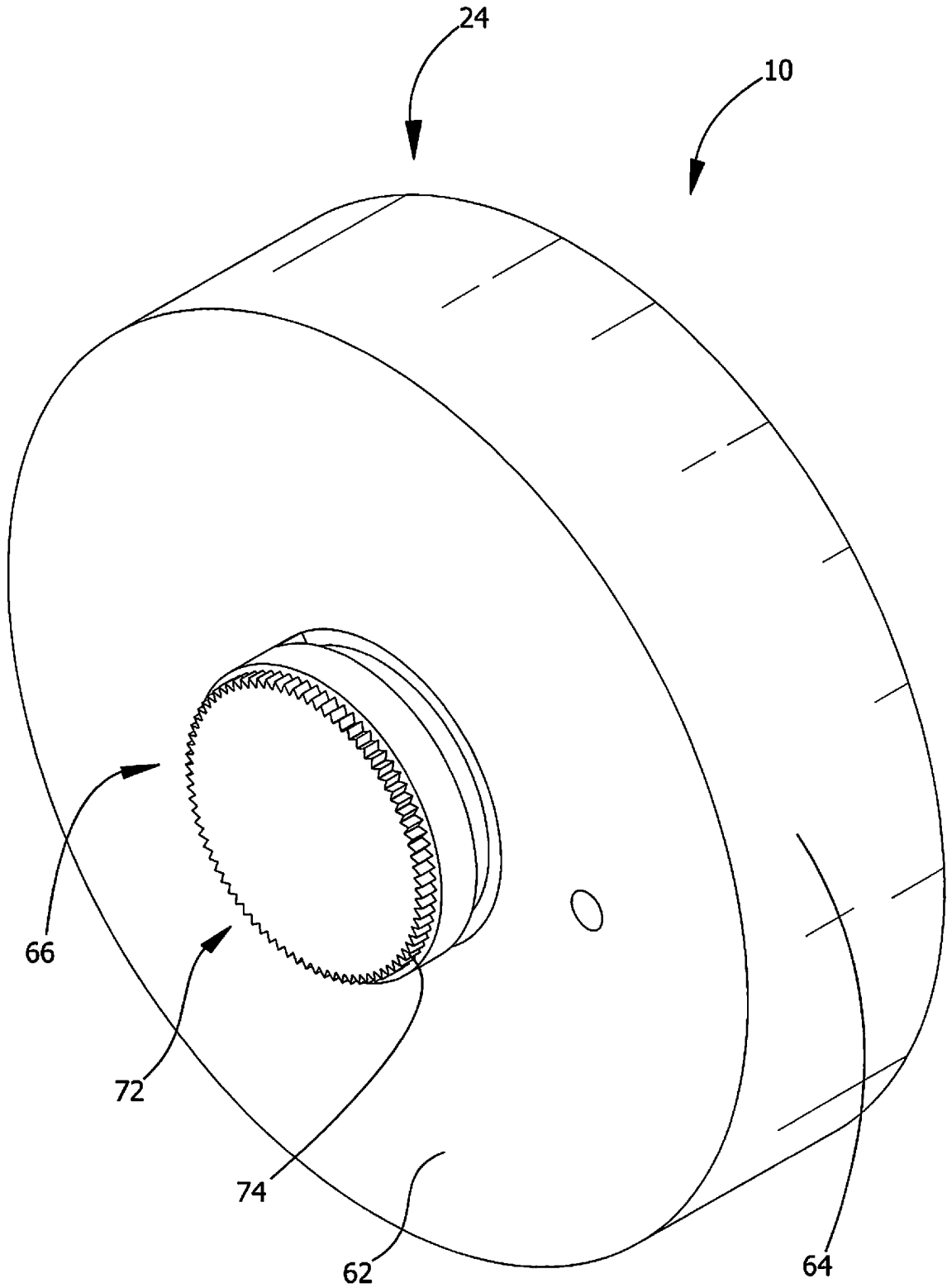


FIG. 4

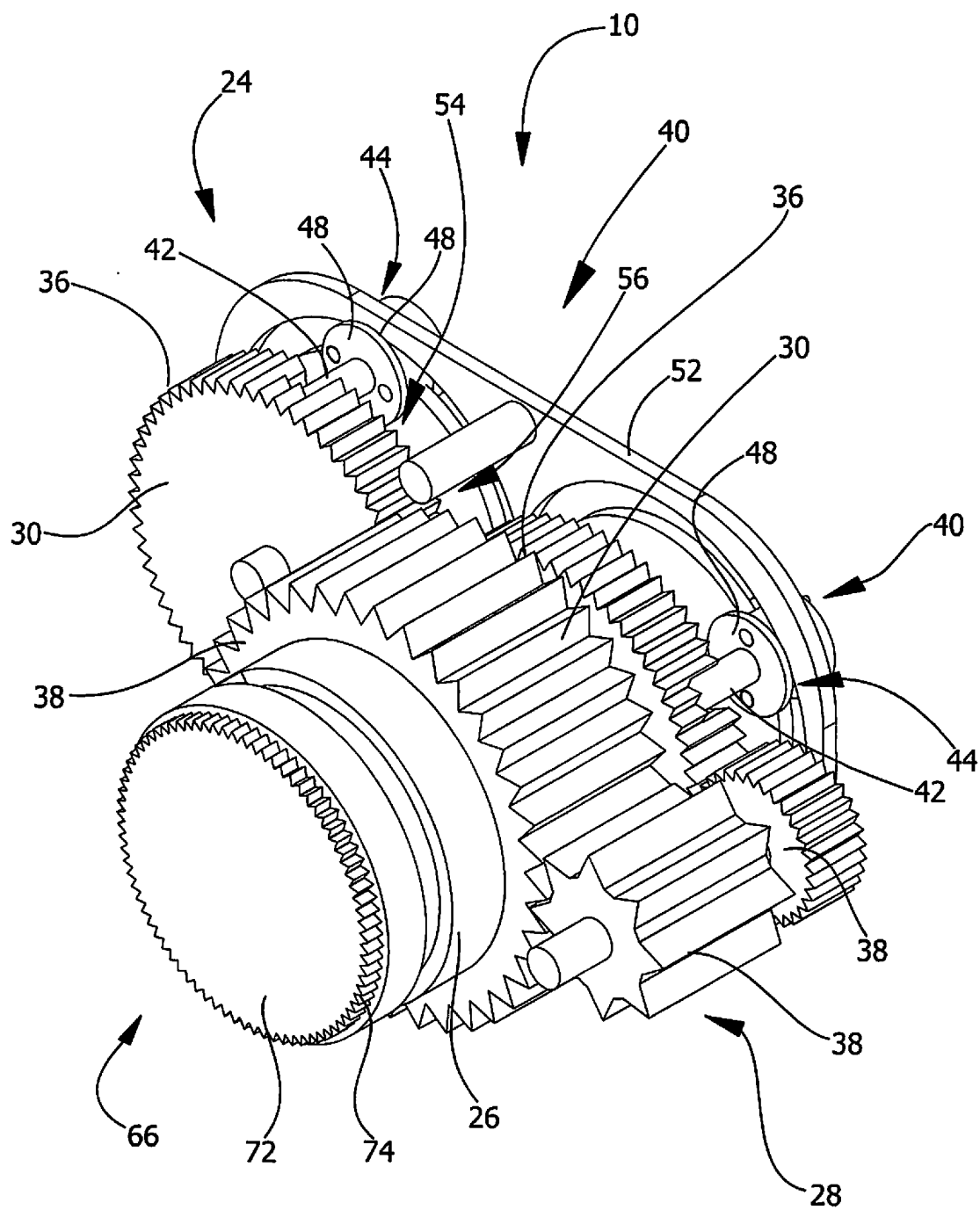


FIG. 5

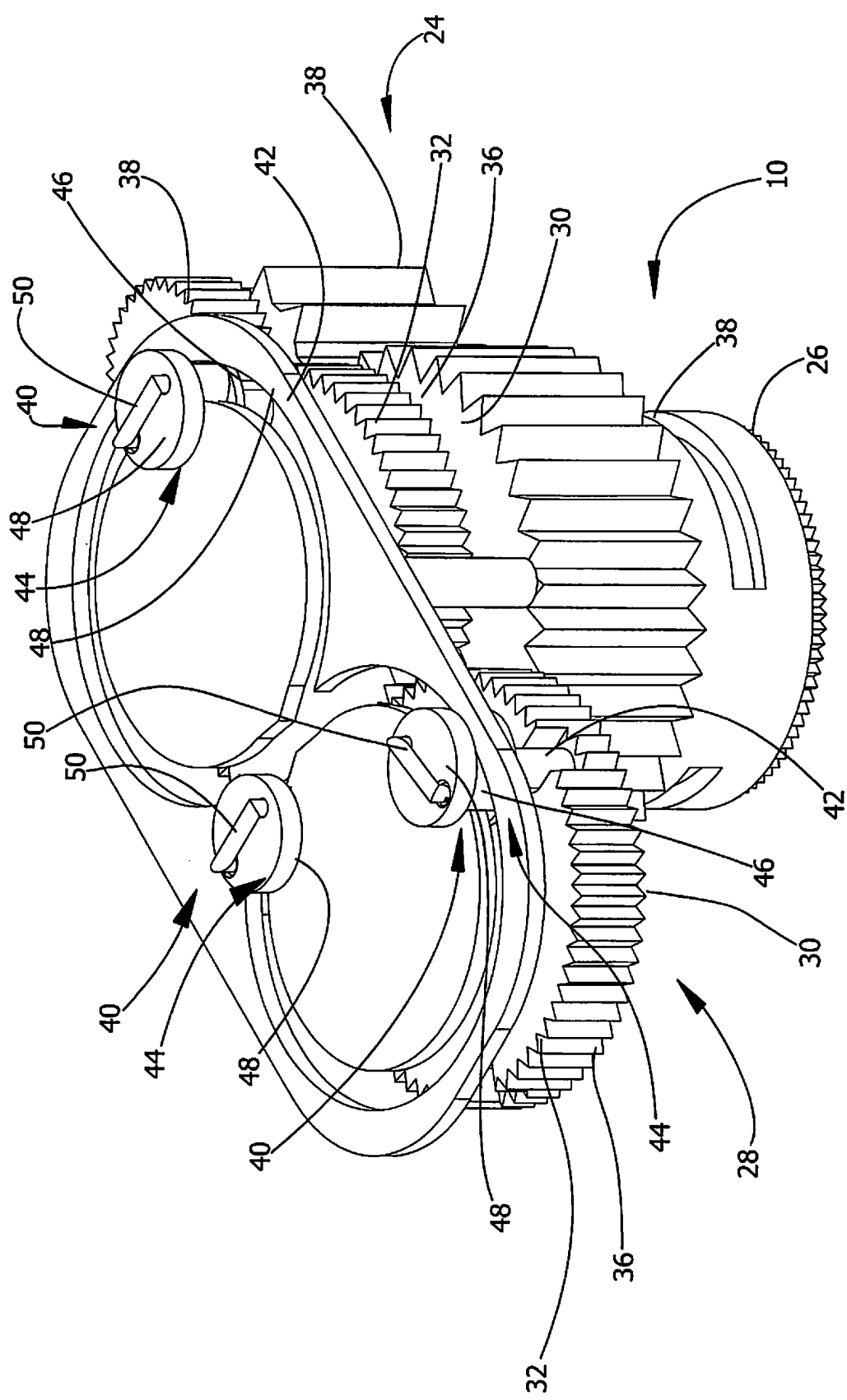


FIG. 6

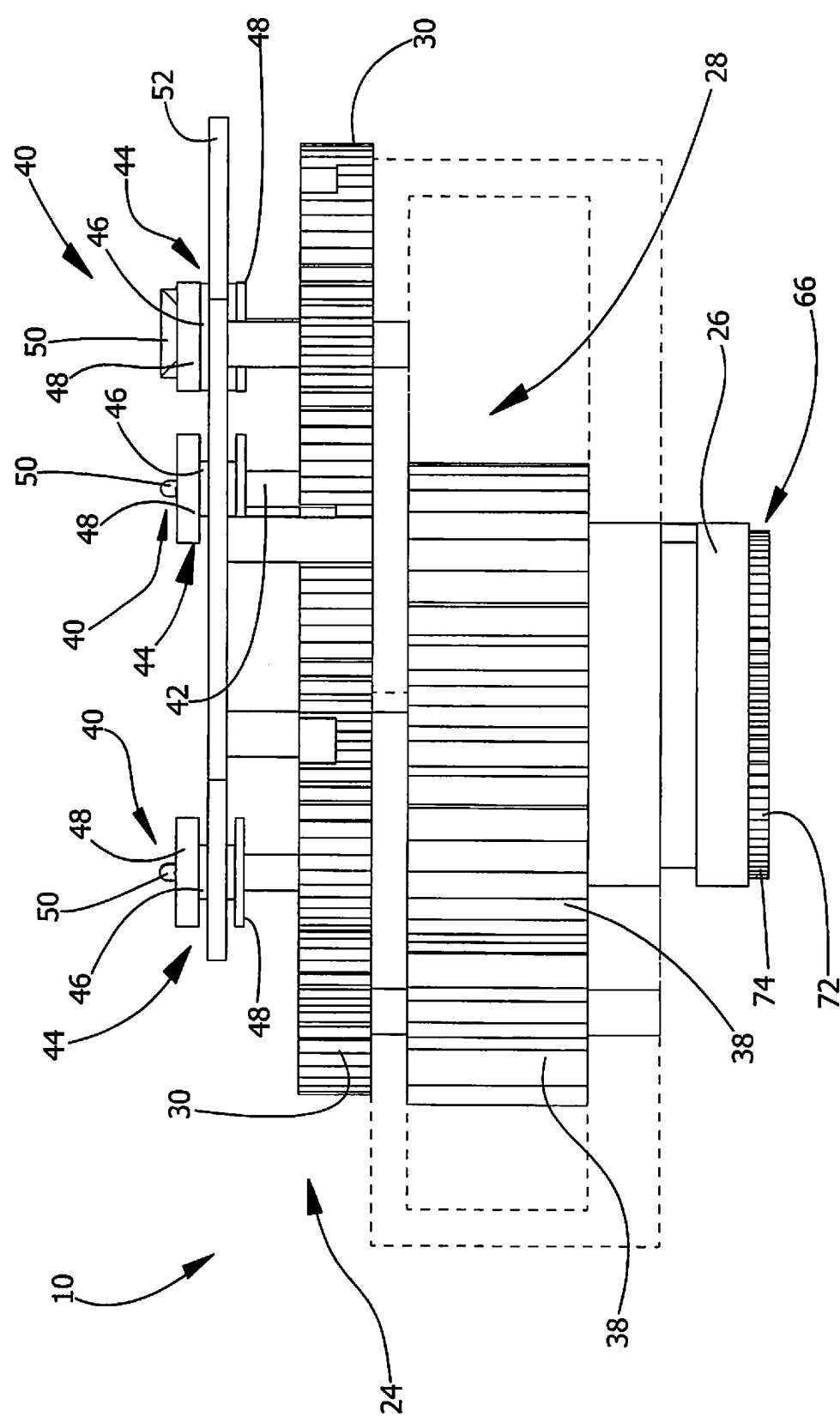


FIG. 7

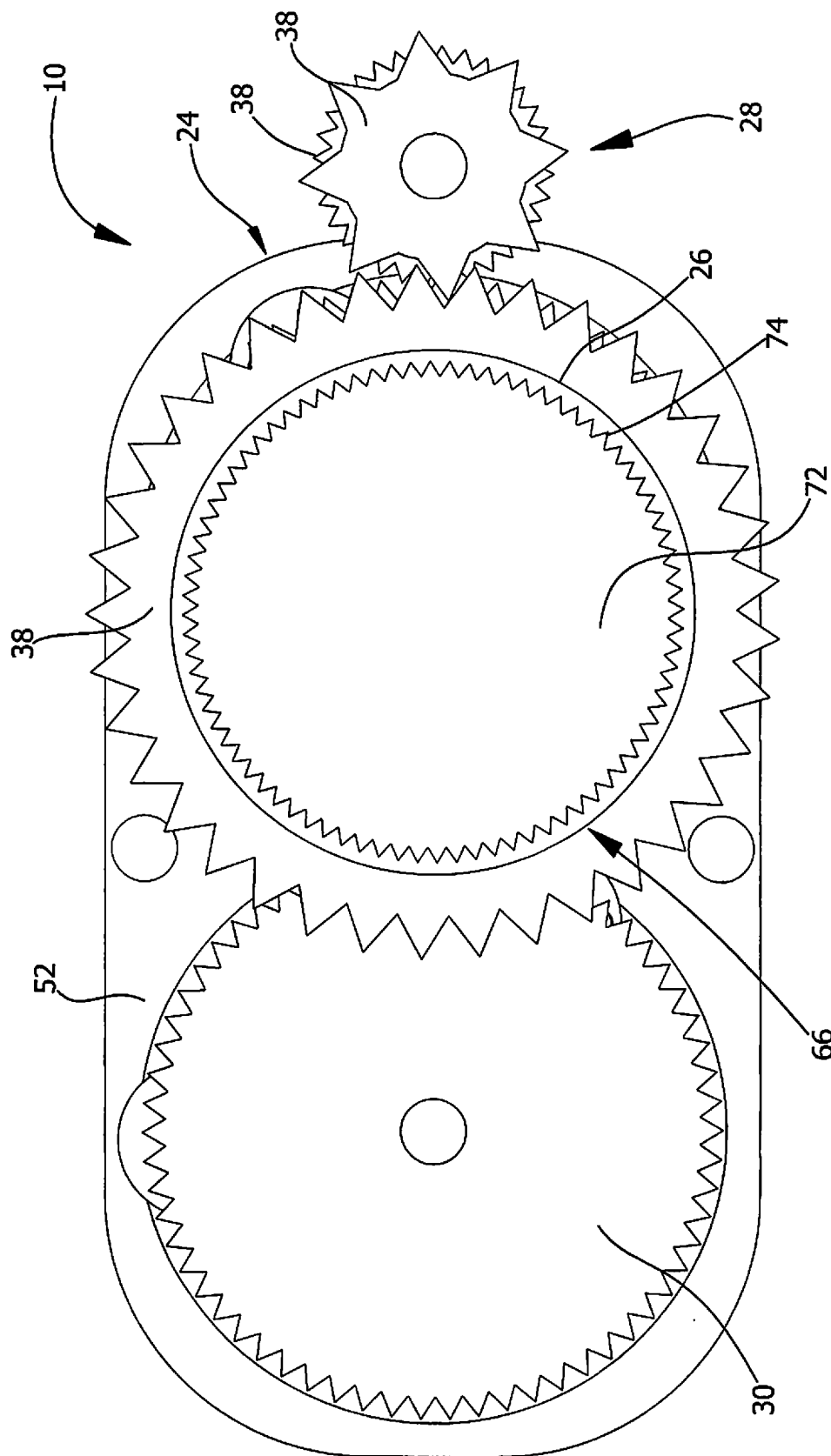


FIG. 8

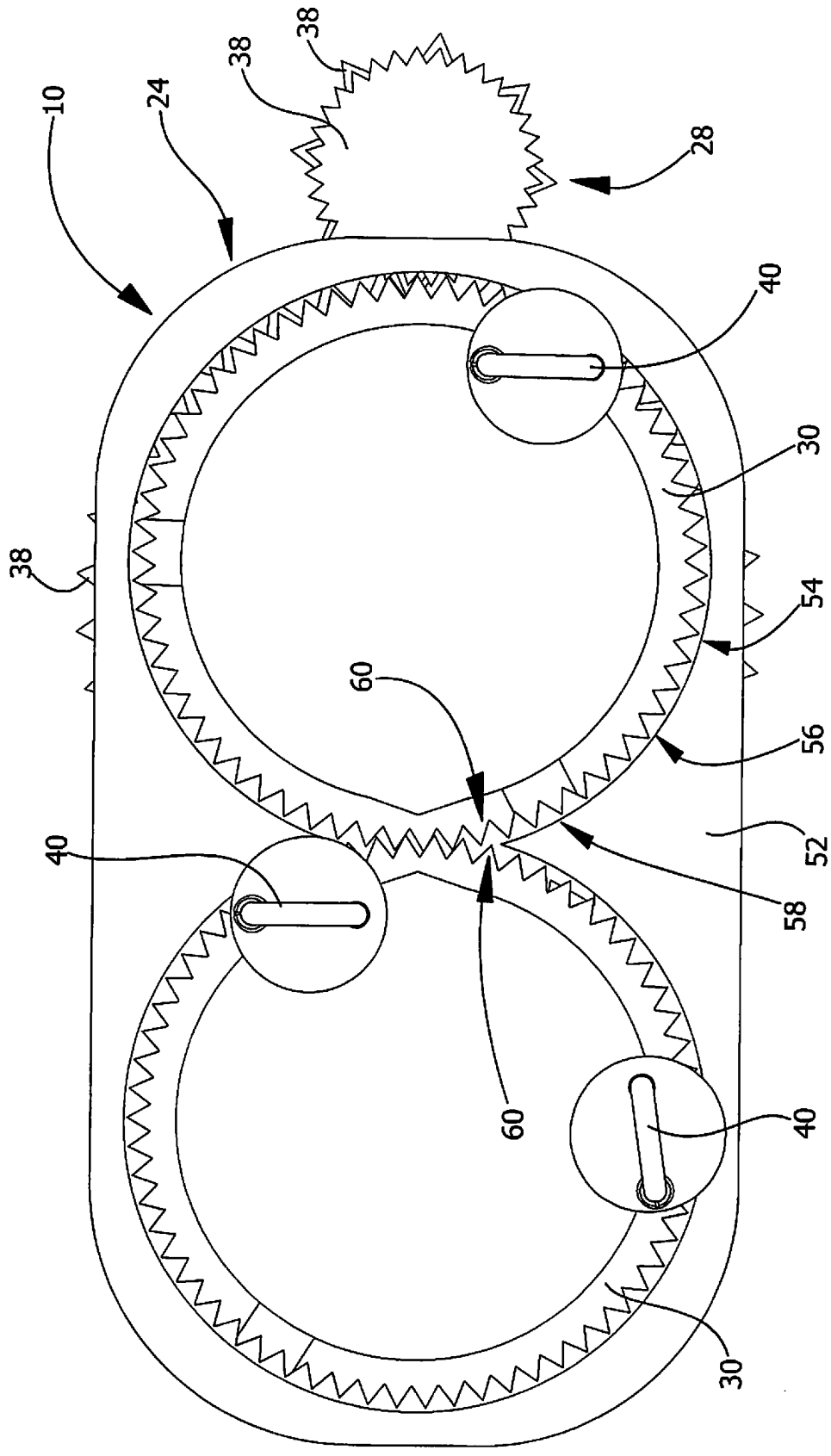
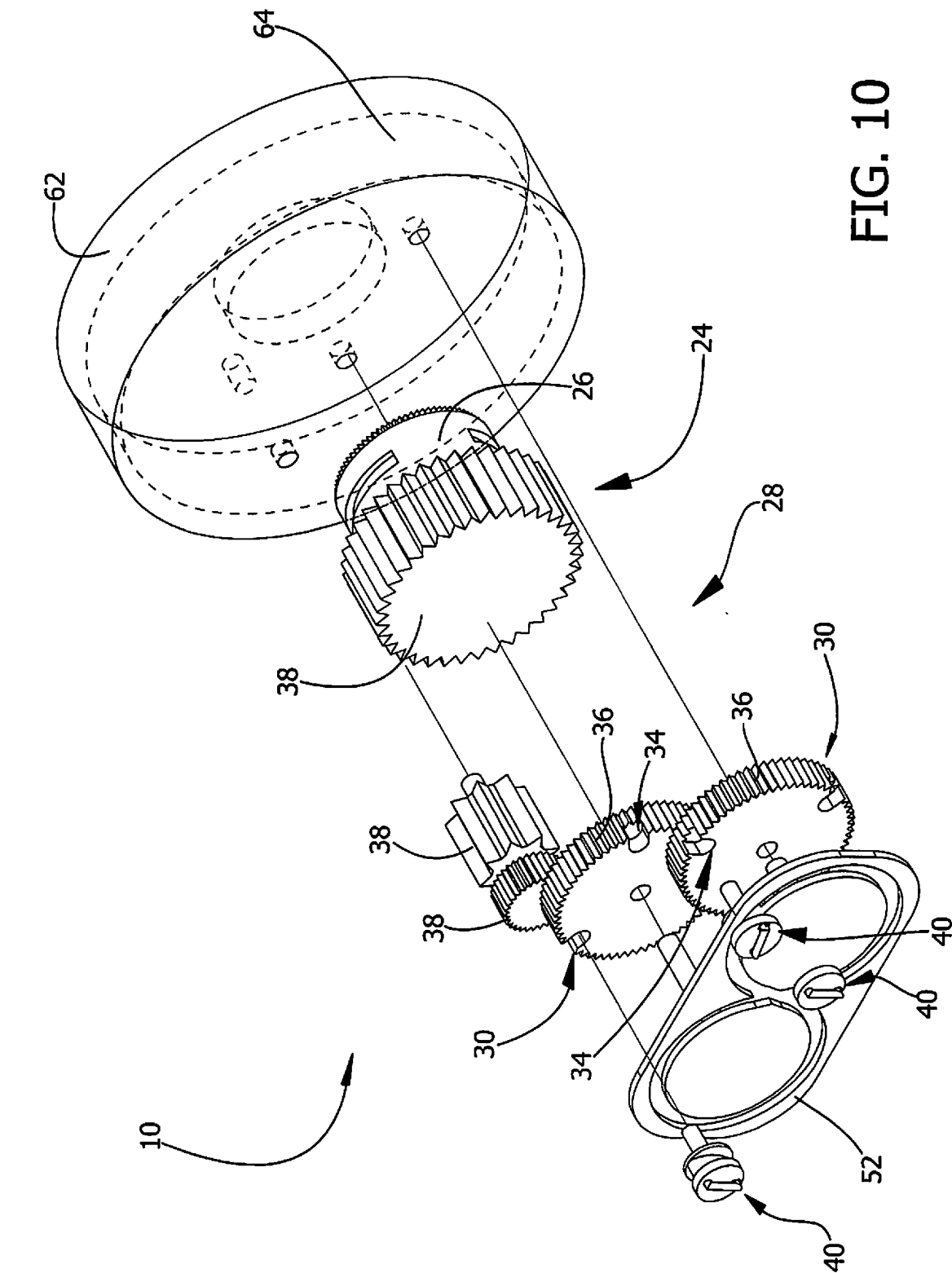


FIG. 9



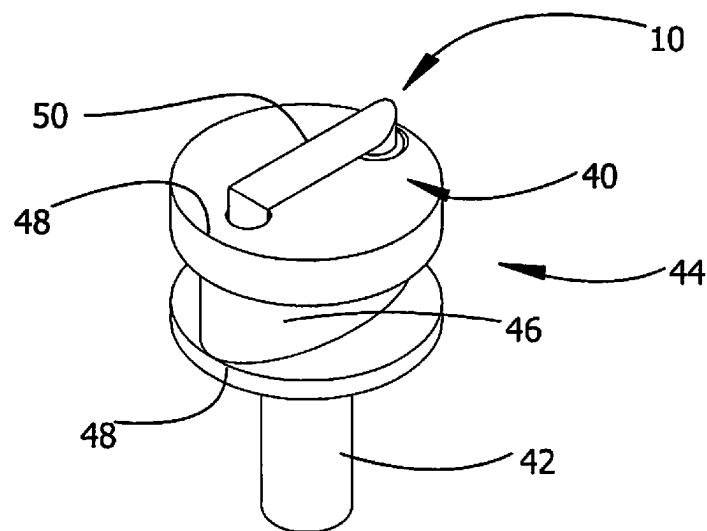


FIG. 11

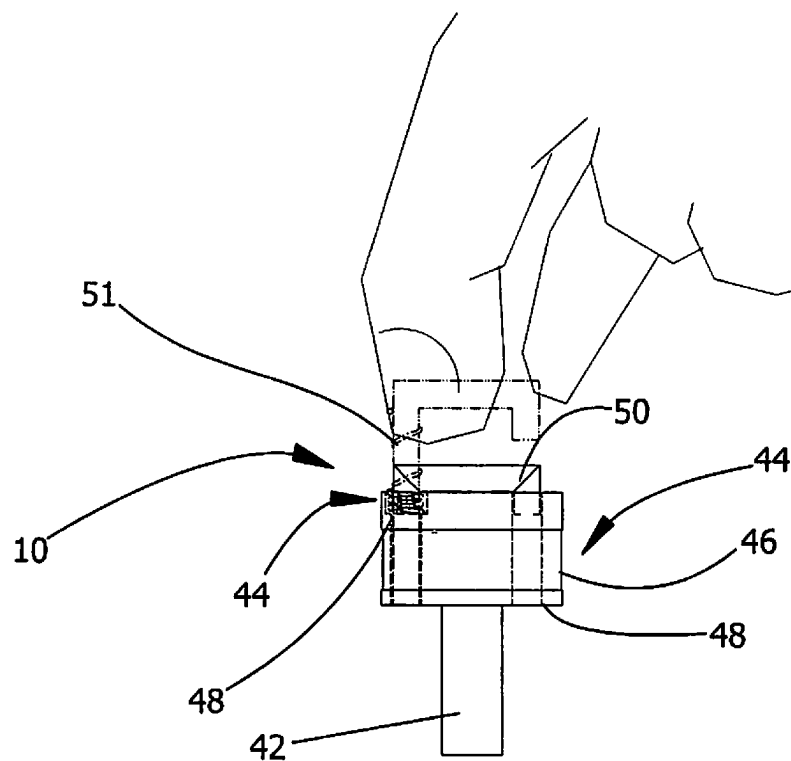


FIG. 12

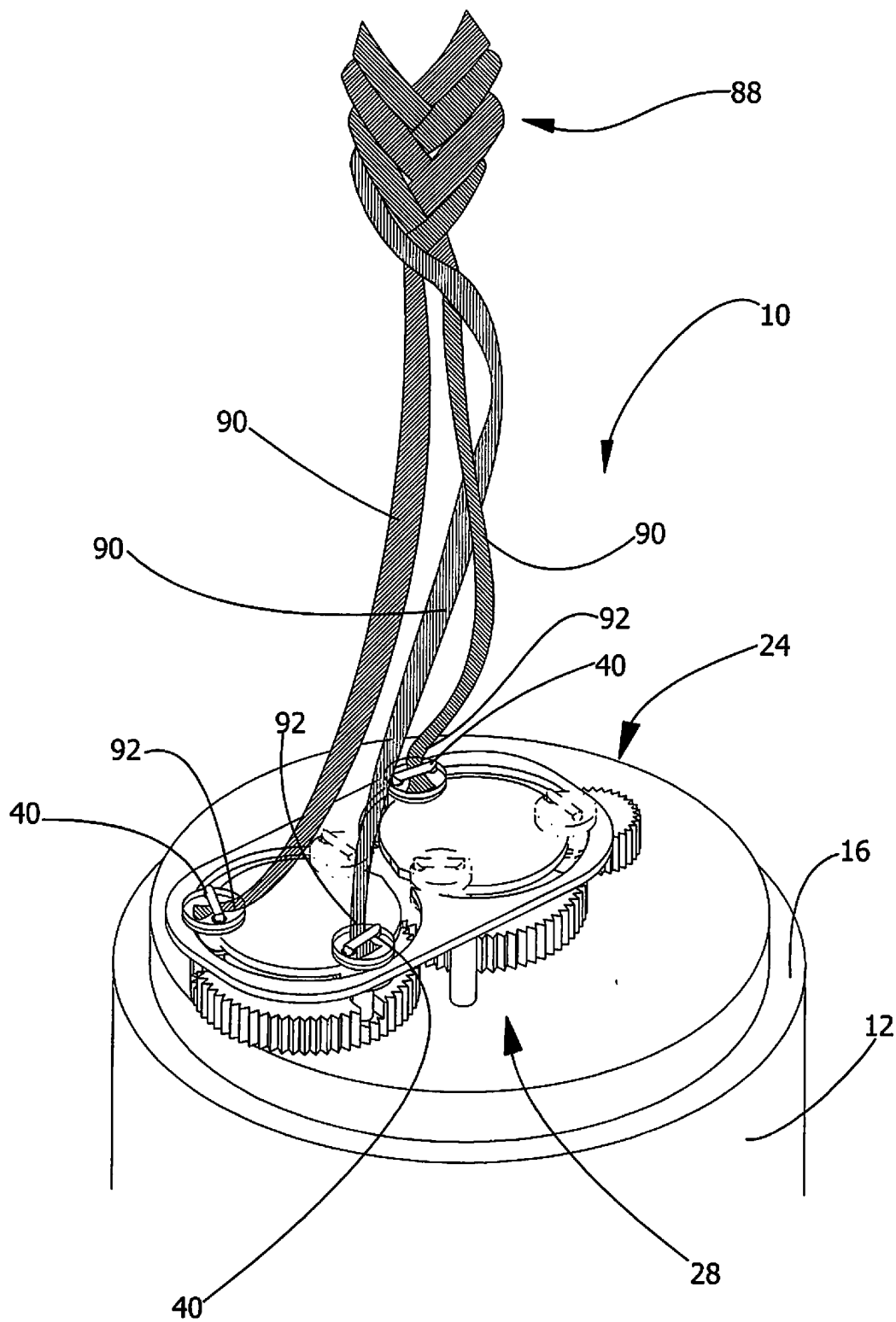


FIG. 13

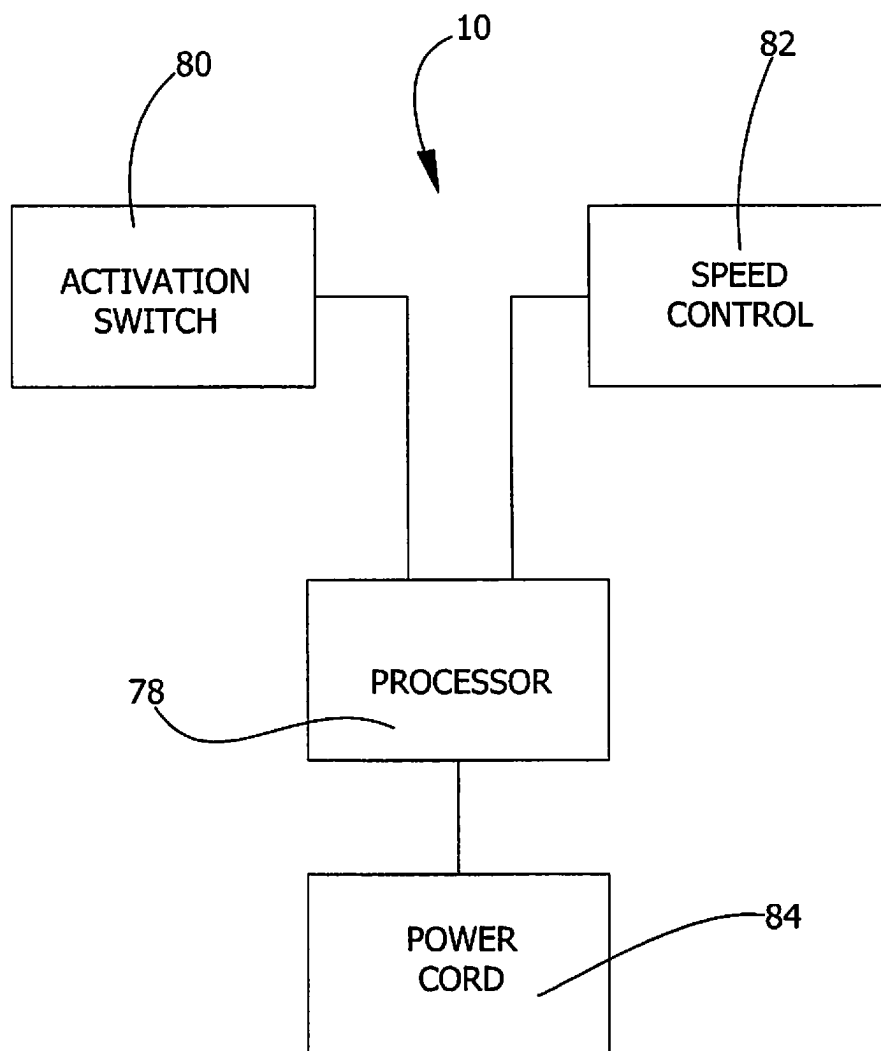


FIG. 14

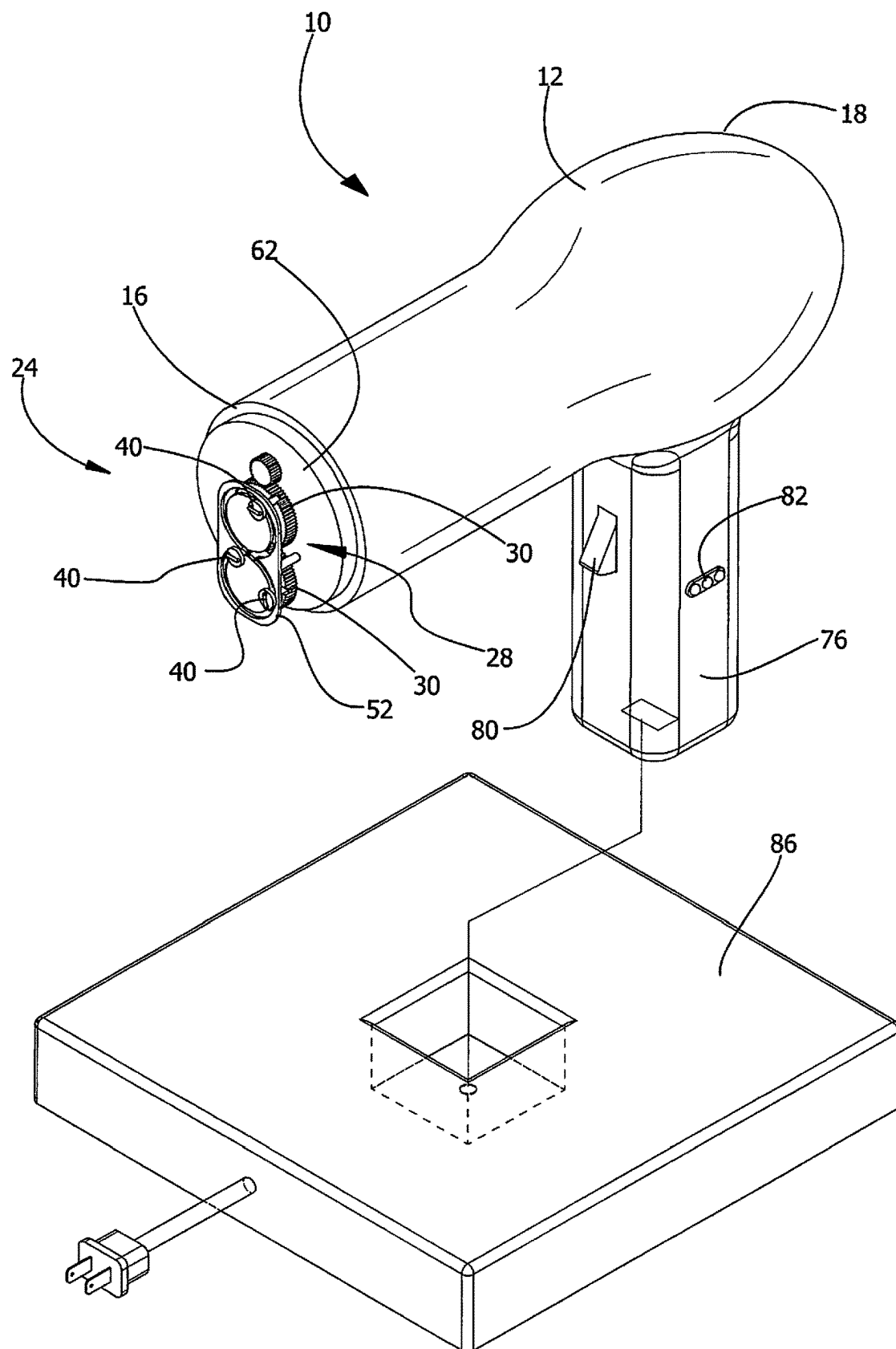


FIG. 15

HAIR BRAIDING DEVICE**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] I hereby claim the benefit under 35 U.S.C., Section 120 of U.S. application Ser. No. 18/220,049 filed on Jul. 10, 2023.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION**(1) Field of the Invention**

[0006] The disclosure relates to hair braiding devices and more particularly pertains to a new hair braiding device for automatically braiding hair.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art relates to hair braiding devices which twist hair to form a braid. Some such devices attach to ends of several tufts of hair via several hair couplers and cause the hair couplers to move in a figure-eight path to form a stronger braid than those in which the tufts of hair are merely twisting in a circular path. The prior art fails, however, to describe such an apparatus which is operable to move the hair couplers in either of a pair of opposing directions along the figure-eight path. Such a function would facilitate the formation of braids in a greater number of styles consisting of alternating braid patterns. The function would also make it possible to operate the device in reverse to untwist a braid. The devices of the prior art also fail to disclose a braiding mechanism which is removable from a motor of the device, which may allow for other attachments, including variations of the braiding mechanism, to be interchangeably placed in engagement with the motor.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a housing which defines an orifice that extends into the housing. A motor is mounted in the housing and is operable to rotate an output shaft of the motor alternately in a first rotational direction

and a second rotational direction opposite from the first rotational direction. A braiding mechanism is operably coupled to the output shaft of the motor and is mounted in the orifice of the housing. The braiding mechanism comprises a plurality of hair couplers which are configured to move along a figure-eight path when the motor is activated. Each hair coupler is configured to attach to a tuft of hair.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. 1 is a perspective view of a hair braiding device according to an embodiment of the disclosure.

[0013] FIG. 2 is an exploded perspective view of an embodiment of the disclosure.

[0014] FIG. 3 is a front perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0015] FIG. 4 is a rear perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0016] FIG. 5 is a rear perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0017] FIG. 6 is a front perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0018] FIG. 7 is a top perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0019] FIG. 8 is a rear detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0020] FIG. 9 is a front detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0021] FIG. 10 is an exploded perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0022] FIG. 11 is a front detail view of a hair coupler of an embodiment of the disclosure.

[0023] FIG. 12 is a side detail view of a hair coupler of an embodiment of the disclosure.

[0024] FIG. 13 is a detail in-use view of an embodiment of the disclosure.

[0025] FIG. 14 is a block diagram of an embodiment of the disclosure.

[0026] FIG. 15 is an alternative embodiment of the disclosure with a charging station for charging a battery.

DETAILED DESCRIPTION OF THE INVENTION

[0027] With reference now to the drawings, and in particular to FIGS. 1 through 15 thereof, a new hair braiding device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral 10 will be described.

[0028] As best illustrated in FIGS. 1 through 15, the hair braiding device 10 generally comprises a housing 12 which defines an orifice 14 that extends into the housing 12. The housing 12 is elongated between a first end 16 and a second end 18, and the orifice 14 is positioned on the first end 16 of the housing 12. A motor 20 is mounted in the housing 12 and is operable to rotate an output shaft 22 of the motor 20 alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction.

[0029] A braiding mechanism 24 is operably coupled to the output shaft 22 of the motor 20. The braiding mechanism 24 is mounted in the orifice 14 of the housing 12 and is removable from the housing 12 and the output shaft 22. The braiding mechanism 24 comprises an input shaft 26, a drivetrain 28, a plurality of hair couplers 40, and a track. The input shaft 26 is coupled to the output shaft 22 of the motor 20 and urges the drivetrain 28 to cause the hair couplers 40 to move in a figure-eight path as guided by the track. The input shaft 26 is axially aligned with the output shaft 22 wherein the output shaft 22 urges the input shaft 26 to rotate with respect to the housing 12 when the output shaft 22 rotates. The drivetrain 28 is operatively coupled to the input shaft 26 and includes a pair of output gears 30 which engage the hair couplers 40.

[0030] The output gears 30 are equal in size to each other and have axes which are oriented parallel to each other. The output gears 30 are also enmeshed with each other such that the pair of output gears 30 rotate in opposite directions from each other. A front side 32 of each output gear 30 of the pair of output gears 30 faces outwardly away from the housing 12. Each output gear 30 of the pair of output gears 30 defines a plurality of cavities 34 in the front side 32 of the output gear 30. The plurality of cavities 34 of each output gear 30 is radially arranged with respect to the output gear 30 and is open at a perimeter 36 of the output gear 30.

[0031] Each hair coupler 40 of the plurality of hair couplers 40 is positioned in one of the cavities 34 of the plurality of cavities 34 of one of the output gears 30 of the pair of output gears 30. Each hair coupler 40 comprises a rod 42, a track engagement member 44, and a hook 50. The rod 42 is positioned in the one cavity 34 and protrudes out of the one cavity 34 beyond the front side 32 of the one output gear 30. The track engagement member 44 is coupled to the rod 42 opposite the one output gear 30, and the hook 50 is coupled to the track engagement member 44 opposite the rod 42. The hook 50 is biased via a tension spring 51 toward the track engagement member 44 in a clamping fashion. The hooks 50 are attachable to hair 88 via clamping the hair between the hooks 50 and the respective track engagement members 44, but the hair couplers 40 may, alternatively or in addition to the hooks 50, eyelets, ties, or the like for such purpose.

[0032] The track engagement member 44 comprises a cam 46 and a pair of shoulders 48. The cam 46 is elongated in an elongation direction which is perpendicular to the rod 42. The cam 46 has an elliptical outer surface with a major axis that extends in the elongation direction, but other elongated

shapes may be suitable. The shoulders 48 are positioned opposite each other across the cam 46.

[0033] The track 52 is spaced forwardly of the pair of output gears 30 and engages the plurality of hair couplers 40. The track 52 defines a groove 54 with a figure-eight shape 56, which the hair couplers 40 are slidable along. The track 52 guides each hair coupler 40 of to move from the one cavity 34 of the one output gear 30 to another cavity 34 of another output gear 28 when the hair coupler 40 crosses through a central portion 58 of the figure-eight shape 56. As depicted most clearly in FIG. 9, the central portion 58 of the figure-eight shape 56 comprises a pair of intersecting straight portions 60. The elongated shape of the cam 46, which is positioned in the groove 54 of the track 52, engages the walls of the groove 54 in one of these intersecting straight portions 60 to be guided across the other intersecting straight portion 60 rather than being allowed to move in a purely circular fashion. The shoulders 48 of the track 52 engagement portion of each hair coupler 40 are positioned opposite each other across the track 52 and act to retain the hair coupler 40 on the track 52.

[0034] The Figures depict three hair couplers 40 corresponding to three cavities 34 in each output gear 30. The cavities 34 of the output gears 30 are arranged such that each cavity 34 of one output gear 30 aligns with one of the cavities 34 of the other output gear 30 when these cavities 34 are positioned near the central portion 58 of the figure-eight shape 56. In this arrangement, each hair coupler 40 may be passed alternately between pairs of aligning cavities 34. Other quantities of hair couplers 40 and cavities 34 are possible and contemplated by the present invention. Other arrangements in which hair couplers 40 are passed through more than two cavities 34 may be used.

[0035] The drivetrain 28 depicted in the Figures shows other gears 38 which operatively couple the input shaft 26 of the drivetrain 28 to the output gears 30. Myriad arrangements of these other gears 38 are possible, including various gear reducer arrangements for causing the output gears 30 to rotate at an output rotational speed which is slower than an input rotational speed of the output shaft 22 of the motor 20. Arrangements in which the output rotational speed is faster than the input rotational speed may also be used. A casing 62 may be used to house or mount one or more gears of the drivetrain 28. The casing 62 may also facilitate inserting and removing the drivetrain 28 into and out of the orifice 14 of the housing 12 respectively by having gripping surfaces ergonomically sized for grasping with a hand. The casing 62 may also facilitate positioning the drivetrain 28 in the orifice 14 by the nature of having a complementary outer surface 64 to a shape of the orifice 14.

[0036] A connector 66 is coupled to the output shaft 22 of the motor 20 and the input shaft 26 of the braiding mechanism 24 and facilitates releasably connecting the braiding mechanism 24 to the motor 20. The connector 66 comprises a first interlocking member 68 coupled to the output shaft 22 and a second interlocking member 72 coupled to the input shaft 26. The first interlocking member 68 has internal teeth 70 which mesh with external teeth 74 of the second interlocking member 72 to couple the output shaft 22 to the input shaft 26.

[0037] A grip 76 is coupled to and extends away from the housing 12. The grip 76 is oriented perpendicularly to the housing 12 and is positioned adjacent to a second end 18 of the housing 12. A processor 78 is operatively coupled to the

motor **20**. An activation switch **80** and a speed control **82** are operatively coupled to the processor **78**. The activation switch **80** is actuatable to selectively activate the motor **20** to rotate the output shaft **22** in the first rotational direction or the second rotational direction. The speed control **82** is actuatable to select a speed of rotation of the motor **20**. The activation switch **80** comprises a trigger positioned on the grip **76**, and the speed control **82** is also mounted to the grip **76**. A power cord **84** is electrically coupled to the processor **78** and is configured to electrically couple to an electrical power source. In other embodiments, a battery or other like power source may be housed in the housing **12** or the grip **76** and may be electrically coupled to the processor **78**. As depicted in FIG. **15**, a charging station **86** may electrically couple to a power source and to the battery to charge the battery in such embodiments.

[0038] In use, ends **92** of tufts **90** of hair **88** are attached respectively to the hooks **50** of the hair couplers **40** by clamping the tufts **90** between the hooks **50** and the track engagement members **44**. The motor **20** is activated in a selected one of the first rotational direction and the second rotational direction and at a selected speed of rotation. The motor **20** causes the hair couplers **40** to move along the figure-eight path, braiding the tufts **90** of hair **88** together. After use of the device **10**, the tufts **90** of hair **88** may be secured in the braided state by a hair tie or the like.

[0039] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0040] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

I claim:

1. A hair braiding device comprising:

- a housing, the housing defining an orifice which extends into the housing;
- a motor mounted in the housing, the motor being operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction; and
- a braiding mechanism being operably coupled to the output shaft of the motor, the braiding mechanism being mounted in the orifice of the housing, the braiding mechanism comprising a plurality of hair couplers being configured to move along a figure-eight path

when the motor is activated, each hair coupler of the plurality of hair couplers being configured to attach to a tuft of hair.

2. The device of claim **1**, wherein the braiding mechanism further comprises:

- an input shaft being coupled to and being axially aligned with the output shaft wherein the output shaft urges the input shaft to rotate with respect to the housing when the output shaft rotates;

- a drivetrain being operatively coupled to the input shaft, the drivetrain including a pair of output gears, the pair of output gears being equal in size and having axes which are oriented parallel to each other, the pair of output gears being enmeshed with each other such that the pair of output gears rotate in opposite directions from each other, a front side of each output gear of the pair of output gears facing outwardly away from the housing, each output gear of the pair of output gears defining a plurality of cavities in the front side of the output gear, the plurality of cavities of each output gear of the pair of output gears being radially arranged with respect to the output gear, each cavity of the plurality of cavities being open at a perimeter of the output gear; wherein each hair coupler of the plurality of hair couplers is positioned in one of the cavities of the plurality of cavities of one of the output gears of the pair of output gears; and

- a track being spaced forwardly of the pair of output gears, the track engaging the plurality of hair couplers, the track being shaped to guide the plurality of hair couplers along the figure-eight path, the track guiding the plurality of hair couplers to be passed between the pair of output gears.

3. The device of claim **2**, wherein each hair coupler further comprises:

- a rod positioned in the one cavity and protruding out of the one cavity beyond the front side of the one output gear;

- a track engagement member being coupled to the rod opposite the one output gear, the track engagement member comprising:

- a cam, the cam being elongated in an elongation direction which is perpendicular to the rod; and

- a pair of shoulders being coupled to the cam, the pair of shoulders being positioned opposite each other across the cam;

wherein the track defines a groove having a figure-eight shape, the plurality of hair couplers being slidable along the groove, the track guiding each hair coupler of the plurality of hair couplers to move from the one cavity of the one output gear to another cavity of another gear when the hair coupler crosses through a central portion of the figure-eight shape, the cam of the track engagement member of each hair coupler of the plurality of hair couplers being positioned in the groove, the pair of shoulders of the track engagement member of each hair coupler of the plurality of hair couplers being positioned opposite each other across the track.

4. The device of claim **3**, wherein the cam has an elliptical outer surface with a major axis extending in the elongation direction.

5. The device of claim **2**, wherein the braiding mechanism is removable from the housing and the output shaft, further

comprising a connector being coupled to the output shaft of the motor and the input shaft of the braiding mechanism, the connector comprising a first interlocking member coupled to the output shaft and a second interlocking member coupled to the input shaft, the first interlocking member and the second interlocking member mating with each other to couple the output shaft to the input shaft.

6. The device of claim 5, wherein the first interlocking member having internal teeth which mesh with external teeth of the second interlocking member to couple the output shaft to the input shaft.

7. The device of claim 1, wherein the housing is elongated between a first end and a second end, the orifice being positioned on the first end of the housing, further comprising a grip being coupled to and extending away from the housing, the grip being oriented perpendicularly to the housing, the grip being positioned adjacent to a second end of the housing.

8. The device of claim 1, further comprising:

- a processor being operatively coupled to the motor;
- an activation switch being operatively coupled to the processor, the activation switch being actuatable to selectively activate the motor to rotate the output shaft in the first rotational direction or the second rotational direction; and
- a speed control being operatively coupled to the processor, the speed control being actuatable to select a speed of rotation of the motor.

9. The device of claim 8, further comprising a power cord being electrically coupled to the processor, the power cord being configured to electrically couple to an electrical power source.

10. The device of claim 1, wherein each hair coupler of said plurality of hair couplers comprises a hook which is coupled to the track engagement member opposite the rod.

11. The device of claim 10, wherein the hook is biased toward the track engagement member in a clamping fashion.

12. A hair braiding device comprising:

- a housing, the housing defining an orifice which extends into the housing, the housing being elongated between a first end and a second end, the orifice being positioned on the first end of the housing;
- a motor mounted in the housing, the motor being operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction;
- a braiding mechanism being operably coupled to the output shaft of the motor, the braiding mechanism being mounted in the orifice of the housing, the braiding mechanism being removable from the housing and the output shaft, the braiding mechanism comprising:
 - an input shaft being coupled to and being axially aligned with the output shaft wherein the output shaft urges the input shaft to rotate with respect to the housing when the output shaft rotates;
- a drivetrain being operatively coupled to the input shaft, the drivetrain including a pair of output gears, the pair of output gears being equal in size and having axes which are oriented parallel to each other, the pair of output gears being enmeshed with each other such that the pair of output gears rotate in opposite directions from each other, a front side of each output gear of the pair of output gears facing outwardly away from the housing, each output gear

of the pair of output gears defining a plurality of cavities in the front side of the output gear, the plurality of cavities of each output gear of the pair of output gears being radially arranged with respect to the output gear, each cavity of the plurality of cavities being open at a perimeter of the output gear;

a plurality of hair couplers, each hair coupler of the plurality of hair couplers being positioned in one of the cavities of the plurality of cavities of one of the output gears of the pair of output gears, each hair coupler comprising:

- a rod positioned in the one cavity and protruding out of the one cavity beyond the front side of the one output gear;

- a track engagement member being coupled to the rod opposite the one output gear, the track engagement member comprising:

- a cam, the cam being elongated in an elongation direction which is perpendicular to the rod, the cam having an elliptical outer surface with a major axis extending in the elongation direction; and

- a pair of shoulders being coupled to the cam, the pair of shoulders being positioned opposite each other across the cam; and

- a hook being coupled to the track engagement member opposite the rod, the hook being biased toward the track engagement member in a clamping fashion; and

- a track being spaced forwardly of the pair of output gears, the track engaging the plurality of hair couplers, the track defining a groove having a figure-eight shape, the plurality of hair couplers being slidable along the groove, the track guiding each hair coupler of the plurality of hair couplers to move from the one cavity of the one output gear to another cavity of another gear when the hair coupler crosses through a central portion of the figure-eight shape, the cam of the track engagement member of each hair coupler of the plurality of hair couplers being positioned in the groove, the pair of shoulders of the track engagement member of each hair coupler of the plurality of hair couplers being positioned opposite each other across the track;

- a connector being coupled to the output shaft of the motor and the input shaft of the braiding mechanism, the connector comprising a first interlocking member coupled to the output shaft and a second interlocking member coupled to the input shaft, the first interlocking member having internal teeth which mesh with external teeth of the second interlocking member to couple the output shaft to the input shaft;

- a grip being coupled to and extending away from the housing, the grip being oriented perpendicularly to the housing, the grip being positioned adjacent to a second end of the housing;

- a processor being operatively coupled to the motor;

- an activation switch being operatively coupled to the processor, the activation switch being actuatable to selectively activate the motor to rotate the output shaft in the first rotational direction or the second rotational direction, the activation switch comprising a trigger positioned on the grip;

a speed control being operatively coupled to the processor, the speed control being actuatable to select a speed of rotation of the motor, the speed control being mounted to the grip; and

a power cord being electrically coupled to the processor, the power cord being configured to electrically couple to an electrical power source.

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